4 **(a)** Two light identical springs are joined in series and a 120 g bar magnet is hung from the lower end as shown in Figure 4.1. The effective spring constant of the combined springs is 11 N m^{-1}.

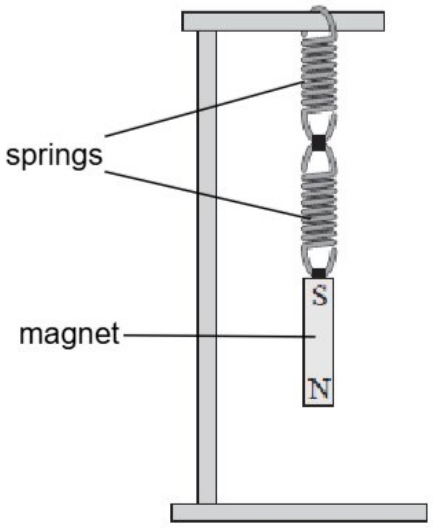


Figure 4.1

The bar magnet is displaced a small distance vertically downward from its equilibrium position and released. It then oscillates in simple harmonic motion.

(i) By considering the forces on the magnet at its equilibrium position and also at a small displacement x downward from its equilibrium position, show that, when released, the magnet's acceleration a is related to its displacement x by the equation:

$$a = -92x$$

Explain your working.

[3]

(ii) Using the expression in **(a)(i)**, calculate the angular frequency of oscillations of the magnet.

angular frequency = ...rad s^{-1} [2]

(iii) The amplitude of the oscillations of the magnet is 3.2 cm. The magnet first passes through the equilibrium position at time $t = 0$.

Calculate the magnitude of the displacement of the magnet from its equilibrium position when $t = 0.40$ s.

displacement = ...cm [2]

(b) Two identical systems shown in Figure 4.1 are now suspended with the North pole of their magnets located inside a coil of wire. The two coils are connected together with conducting wires as shown in Figure 4.2.

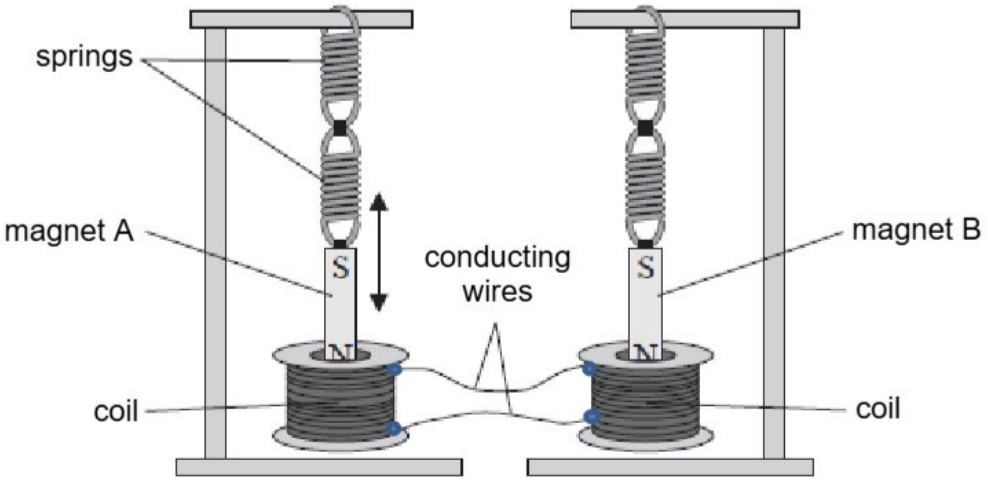


Figure 4.2

Magnet A is displaced so that it oscillates vertically. The North pole of magnet A moves into and out of the coil as it oscillates. As this motion continues, magnet B starts to oscillate. The amplitude of the oscillations of magnet B increases over time.

Explain why magnet B starts to oscillate and why the amplitude of oscillations increases.

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[Total: 11]